

TCS349: Responsible and Explainable AI

Unit 1 | Lectures 6 & 7

Measuring Bias & Defining Fairness in AI

From subgroup analysis and confusion-matrix measures that detect bias, to the formal group and individual notions of fairness that define what a fair outcome should look like.

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Subgroup Analysis



Confusion-Matrix Metrics



Group Fairness



Individual Fairness

1. What is Subgroup Analysis?

Subgroup analysis evaluates model performance separately for defined groups - such as gender, age or region - instead of relying on one overall average.

A model can look highly accurate overall while performing poorly for a particular subgroup.

Groups are usually defined using protected or sensitive attributes relevant to the application.

The goal is to surface disparities in accuracy, error rates or outcomes before a model is deployed.

2. Confusion Matrix Recap

Every subgroup-wise bias measure is built from one tool: the confusion matrix of a binary classifier (e.g., loan approved / not approved).

	Predicted: Positive	Predicted: Negative
Actual: Positive	TP - True Positive	FN - False Negative
Actual: Negative	FP - False Positive	TN - True Negative

3. Confusion-Matrix-Based Bias Measures

False Positive Rate $FPR = FP / (FP + TN)$ - how often the model wrongly predicts the positive outcome.

False Negative Rate $FNR = FN / (FN + TP)$ - how often the model misses a true positive.

True Positive Rate $TPR = TP / (TP + FN)$ - also called recall or sensitivity.

A large gap in these rates between subgroups signals potential bias, even when overall accuracy looks high.

4. Worked Example - Scenario Setup

A bank's AI model predicts loan approval. We evaluate it on 200 test applicants split into two demographic groups of 100 each, with 40 genuinely creditworthy applicants per group.

Group	Sample Size	Actually Creditworthy	TP	FP	FN	TN
Group A	100	40	32	12	8	48
Group B	100	40	20	12	20	48

5. Worked Example - Computing the Metrics

Using the confusion-matrix values from the previous slide:

Group	Sample Size	Actually Creditworthy	TP	FP	FN	TN
Group A	100	40	32	12	8	48
Group B	100	40	20	12	20	48

Metric	Group A	Group B	Gap
TPR (Recall)	$32/40 = 80\%$	$20/40 = 50\%$	30 pts
FPR	$12/60 = 20\%$	$12/60 = 20\%$	0 pts
FNR	$8/40 = 20\%$	$20/40 = 50\%$	30 pts

6. Interpreting Bias from the Numbers

Both groups share the same FPR (20%) - but TPR and FNR differ sharply between them.

A creditworthy Group B applicant is far more likely to be wrongly denied a loan: FNR of 50% versus 20% for Group A.

Equal FPR alone does not guarantee a fair outcome - the TPR/FNR gap reveals unequal opportunity.

This numeric gap is exactly what the group-fairness definitions in the next part are designed to formalise and measure.

7. Tools for Bias Detection

Fairlearn (Microsoft): a Python library for assessing and mitigating fairness issues in ML models.

IBM AI Fairness 360 (AIF360): an open-source toolkit offering 70+ fairness metrics and bias-mitigation algorithms.

Aequitas: an audit toolkit that generates subgroup-wise bias reports directly from model predictions.

These tools automate the same subgroup and confusion-matrix analysis performed manually in this lecture.

8. From Detecting Bias to Defining Fairness

Detecting a numeric gap tells us bias exists - it does not tell us what a "fair" outcome should look like.

Fairness in AI is the formal, mathematical attempt to define equitable treatment across individuals and groups.

There is no single universal definition of fairness - the appropriate notion depends on context and stakes.

The rest of this session introduces the two broad families: group fairness and individual fairness.

9. Group Fairness - Demographic Parity

Group fairness checks whether a chosen statistical measure is equal across defined subgroups.

Demographic parity (statistical parity): the positive-prediction rate should be equal across groups, regardless of actual outcomes.

Formally: $P(\text{prediction} = \text{positive} \mid \text{Group A}) = P(\text{prediction} = \text{positive} \mid \text{Group B})$.

Limitation: demographic parity can conflict with accuracy when groups genuinely differ in their base rates.

10. Equal Opportunity & Equalized Odds

Equal opportunity: TPR should be equal across groups - qualified individuals get an equal chance of a positive outcome.

Equalized odds: both TPR and FPR should be equal across groups - a stricter condition than equal opportunity alone.

Both definitions directly reuse the confusion-matrix measures computed earlier in this session.

A deeper look at these metrics and their trade-offs follows in Lecture 8.

11. Individual Fairness

Individual fairness requires that similar individuals receive similar predictions, regardless of group membership.

Fairness is defined at the level of a person, not a demographic category.

It depends on a similarity metric - deciding which individuals count as "similar" is itself a design choice.
Individual fairness can conflict with group fairness even when both are pursued in good faith.

12. Choosing the Right Fairness Notion

Dimension	Group Fairness	Individual Fairness
Focus	Subgroups defined by sensitive attributes	Similarity between individuals
Key Question	Are outcomes balanced across groups?	Are similar people treated similarly?
Common Metrics	Demographic parity, equal opportunity, equalized odds	Similarity-based consistency
Main Risk	May still treat some individuals within a group unfairly	May allow group-level disparities to persist

Self-Learning & Team Activity (Unit I)

Self-Learning Activity: Review a real AI failure and identify the ethical issue, source of bias, affected stakeholders and relevant fairness concerns.

Team Activity: Analyse an AI application used in daily life and propose measures to make it fairer and safer.

Suggested reading: Denis Rothman, "Hands-On Explainable AI (XAI) with Python" (Packt, 2020); Fairlearn and AIF360 documentation. Link: <https://aif360.readthedocs.io/en/stable/>